



CANDIDATE
NAME

CG

INDEX NO

COMPUTING

Paper 2 Practical

9569/02

2026

1 hour 15 minutes

Additional Materials:

Removable storage device with the following files:

- `PROMO_template.ipynb`
- `SCORES.TXT`
- `Database.db`

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.

All tasks must be done in the computer laboratory. You are not allowed to bring in or take out any pieces of work or materials on paper or electronic media or in any other form.

Approved calculators are allowed.

Save each task as it is completed.

The use of built-in functions, where appropriate, is allowed for this paper unless stated otherwise.

The number of marks is given in brackets [] at the end of each task.

The total number of marks for this paper is **40**.

Instruction to candidates:

Your program code and output for all the questions should be saved in a single file and downloaded as

PROMO_<your class>_<your name>.ipynb

For each of the sub-tasks, add a comment statement, at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 1.1  
Program Code
```

Output:

In [2]:

```
#Task 1.2  
Program Code
```

Output:

In [3]:

```
#Task 1.3  
Program Code
```

Output:

- 1 A student intends to deposit \$ x at the beginning of every month into a bank account where $10 \leq x \leq 100$. The bank will pay a compound interest of 2% per month on the last day of each month.

Task 1.1

Write program code to:

- prompt the student to enter the amount x he intends to deposit
- validate that the input value is between 10 and 100, both inclusive
- if the input value is not between 10 and 100, print an error message "Invalid Input" and prompt the student to input the value again
- repeat until a valid input is obtained. [2]

Task 1.2

Write program code for the function `calculate(n)` to calculate the amount of money in the bank account at the end of n months, correct to 2 decimal places. [2]

Task 1.3

Write program code for the function `months(x)` to calculate the least number of months needed for the student to save at least \$2000, given that the student deposit \$ x at the beginning of every month.

The function should return a tuple containing the least number of months required and the amount of money in the bank account at the end, correct to 2 decimal places. [4]

- 2 A text file, SCORES.TXT, contains the results of the students for their recent examination.

Each line contains the student's name, Civics Group (CG) and the scores for English, Mathematics and Science, separated by commas.

Task 2.1

Write program code to:

- read the information from the text file and return a list of tuples
- each tuple contains the student's data in the following format:

```
(Name, CG, English_score, Mathematics_score, Science_score)
```

[6]

- 3 A research team from a university is studying the temperature variations within a day at a specific location.

An example of the hourly temperature readings for a particular day is as follows:

```
temp_data = [('18-Sep', '1:00AM', '24'),  
             ('18-Sep', '2:00AM', '24'),  
             ('18-Sep', '3:00AM', '24'),  
             ...  
             ('18-Sep', '10:00PM', '28'),  
             ('18-Sep', '11:00PM', '28'),  
             ('18-Sep', '12:00AM', '26')]
```

Task 3.1

Write program code for the helper function `get_temp(tup)` to return the temperature value of a particular tuple `tup` as an integer.

[1]

Task 3.2

Write program code for the function `insertion_sort(lst)` to implement the insertion sort algorithm to arrange the hourly temperature readings in the list `lst` in ascending order.

Test your program with the list of hourly temperature readings provided in the template.

[5]

- 4 The distance for the inner most lane in a running track is 400.00 m. The distance for the lane next to it is 7.67 m longer and every subsequent adjacent lane is also 7.67 m longer.

Task 4

Write a recursive program code for the function `distance(n)` to compute and return the distance of the n th lane.

Use your program code to calculate the distance of the 8th lane in a running track.

[4]

- 5 The mail server at YIJC uses a hash table of size 20 to store the email addresses of all the users.

For example, the email address for the user 'abc' is `abc@yijc.edu.sg` and the sum of the ASCII values of the letters in the username is calculated as follow:

$$\text{ord}('a') + \text{ord}('b') + \text{ord}('c') = 97 + 98 + 99 = 294$$

Therefore, the hash value

$$\text{hash}('abc@yijc.edu.sg') = 294 \% 20 = 14$$

Task 5.1

Write program code for the function `hash(email)` to compute and return the hash value of the username.

[3]

The hash value computed for the username is used as the index position to store the email address in the hash table. The following list of email addresses are to be stored in a hash table:

```
email_list = ['mrs_ho@yijc.edu.sg' , 'mdm_goh@yijc.edu.sg',  
              'mr_lee@yijc.edu.sg', 'mr_yeo@yijc.edu.sg']
```

Task 5.2

Write program code for the function `hashtable(lst)` to:

- initialise an empty hash table of size 20
- store all the emails in the list `lst` into the hash table according to the index positions found using the hash values of the usernames
- resolve any collision using the linear probing strategy
- return the hash table

Test your program code with the given list `email_list`.

[6]

6 The following is the first paragraph of a fantasy story:

```
paragraph = "In a land far away, there was a legendary kingdom
              known as Aurum, where gold, silver, and diamonds
              shimmered in every corner. The palace walls gleamed
              with golden bricks, while the streets were paved
              with silver stones that sparkled under the sun. The
              king, wore a crown decorated with diamonds. His
              throne, carved from pure gold, was said to radiate
              power and fortune to all who sat upon it."
```

A treasure hunter is searching for gold, silver and diamond in this fantasy land. He uses a dictionary data structure to store all the treasures he found.

Task 6.1

Write program code for the function `search(para)` to:

- search through the passage `para` looking for the word "gold", "silver" and "diamonds"
- count the number of times each of these words are found in the passage and store them in a dictionary
- return the dictionary.

Test your program code with the passage `paragraph`. Do not miss "gold" in the word "golden".

[4]

After stealing all the treasures found, the hunter casts a spell to convert them into wood.

Task 6.2

Write program code for the function `cast_spell(para)` to:

- change all the word "gold", "silver" and "diamonds" found in the passage `para` to "wood"
- return the modified passage. [3]

END OF PAPER 2

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CANDIDATE
NAME

CG

INDEX NO

COMPUTING

Paper 1 Written

9569/01

2026

45 minutes

Additional Materials: 4-page Answer booklet

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with the question paper.
You should follow the instructions on the front cover of the answer booklet.
If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.
Approved calculators are allowed.

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is **25**.

This document consists of **4** printed pages.

1

Index	0	1	2	3	4	5	6	7
arr	33	37	42	51	62	71	80	83

(a) Describe using the binary search algorithm to find the value 42 in the array `arr`. [3]

(b) A student claims that both the linear search and binary search algorithms are equally efficient because they require the same number of steps to find the value 42.

Explain whether the claim is true. [4]

2 The following is an unsorted array:

`array = [5, 2, 7, 1, 3, 8, 6, 4]`

(a) With the aid of a diagram, describe how the elements can be arranged in ascending order using the merge sort algorithm. [4]

The elements in the unsorted array are inserted into a binary search tree (BST) in the order of the index number.

(b) Draw the resulting BST. [3]

3 RGB (red, green and blue) is a system of representing the colours for digital lighting devices. Red, green and blue can be combined in various proportions to obtain any colour in the visible spectrum.

In the following table, a colour is represented with two hexadecimal digits each to represent the red, green and blue components, and each colour component has a value ranging from 00 to FF:

Colour	RGB Colour (Hexadecimal Representation)
Black	00:00:00
White	FF:FF:FF
Red	FF:00:00
Green	00:FF:00
Blue	00:00:FF
Violet	EE:82:EE
SeaGreen	2E:8B:57
Chocolate	D2:69:1E
MidnightBlue	19:19:70

A student is writing a program to remotely control the LED lightings on a drone. The communication requires the data to be in binary numbers. Three sets of 8-bit binary data will be used to represent the red, green and blue colour components.

- (a) With clear working, show the steps required to convert the hexadecimal representation for the `Chocolate` colour to a binary representation. [2]

- (b) State and explain two advantages of using the hexadecimal representation for numbers in computer systems, particularly when working with digital data, as compared to using the decimal representation. [2]

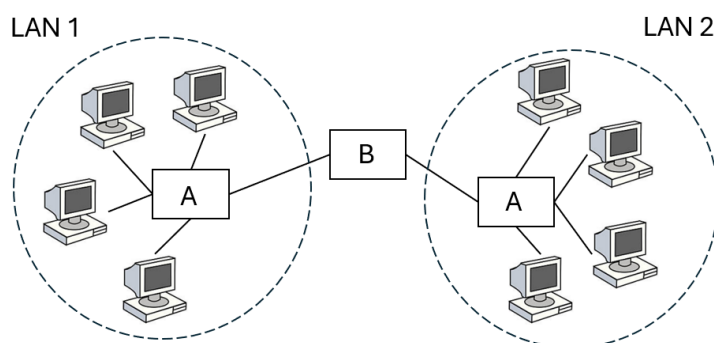
- 4 The following database table `OrderDetails` is used to track customers' purchases:

CustomerID	ProductID	Quantity	UnitPrice	Date
C001	P2001	2	10.00	2024-09-10
C001	P2002	1	15.00	2024-09-10
C001	P2003	3	7.50	2024-09-11
C002	P2001	1	10.00	2024-09-12
C002	P2004	5	8.00	2024-09-12
C003	P2002	2	15.00	2024-09-13
C003	P2005	1	20.00	2024-09-13
C004	P2003	4	7.50	2024-09-14
C004	P2003	1	7.50	2024-09-15

- (a) Suggest a suitable Primary Key (PK) for the table `OrderDetails`. [1]

- (b) Explain two functions of a PK for a table in a relational database. [2]

- 5 Two companies connected their local area network (LAN) using network cables and devices.



- (a) Name the network devices A and B used to connect the computers within a LAN and across the LANs. [2]

- (b) State two main differences between the devices A and B. [2]

END OF PAPER 1